

Exercise (4.1.1 - Stefano Fochesatto). Let G act on the set A . Prove that if $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = gG_ag^{-1}$ (G_a is the stabilizer of a). Deduce that if G acts transitively on A then the kernel of the action is $\bigcap_{g \in G} gG_ag^{-1}$.

Proof: Let G act on the set A . Suppose $a, b \in A$, and $b = g \cdot a$ for some $g \in G$. Let $g_1 \in G_b$ and consider $g^{-1}g_1g \cdot a$. Note that since $g \cdot a = b$ and $g_1 \in G_b$ we get that $g^{-1}g_1(g \cdot a) = g^{-1}g_1 \cdot b = g^{-1}b = a$. It follows that $g^{-1}g_1g \in G_a$, therefore $g_1 \in gG_ag^{-1}$ and thus $G_b \subseteq gG_ag^{-1}$. Let $g_1 \in gG_ag^{-1}$, therefore $g_1 = gg_ag^{-1}$ where $g_a \in G_a$. Consider $g_1 \cdot b$ and note that since $g \cdot a = b$ and $g_a \in G_a$ we get $g_1 \cdot b = gg_a(g^{-1} \cdot b) = gg_a \cdot a = g \cdot a = b$. It follows that $g_1 \in G_b$ and therefore $gG_ag^{-1} \subseteq G_b$ thus $G_b = gG_ag^{-1}$.

Suppose G acts transitively on A . Recall that the kernel of an action is the set of all $g \in G$ that map everything in A to itself, and therefore must be the intersection of all the stabilizers. Fix $a \in A$, and note that since G acts transitively on A for all $b \in A$ there exists a $g \in G$ such that $g \cdot a = b$ thus by the previous result we get,

$$\bigcap_{b \in A} G_b = \bigcap_{g \in G} gG_ag^{-1}$$

□

Exercise (4.2.2 - Stefano Fochesatto). List the elements of S_3 as 1, (12), (23), (13), (123), (132) and label these with the integers 1, 2, 3, 4, 5, 6 respectively. Exhibit the image of each element of S_3 under the left regular representation of S_3 into S_6 .

Solution: Note that $S_3 = \langle (12), (23) \rangle$. First let's compute the image of (12) under the left regular representation of S_3 ,

$$(12)1 = (12) = \sigma_{(12)}(1) = 2$$

$$(12)(12) = 1 = \sigma_{(12)}(2) = 1$$

$$(12)(23) = (132) = \sigma_{(12)}(3) = 6$$

$$(12)(13) = (123) = \sigma_{(12)}(4) = 5$$

$$(12)(123) = (13) = \sigma_{(12)}(5) = 4$$

$$(12)(132) = (23) = \sigma_{(12)}(6) = 3.$$

Thus $\sigma_{(12)} = (12)(45)(36)$. Computing the image of (23) under the left regular representation of S_3 ,

$$(23)1 = (23) = \sigma_{(23)}(1) = 3$$

$$(23)(12) = (123) = \sigma_{(23)}(2) = 5$$

$$(23)(23) = 1 = \sigma_{(23)}(3) = 1$$

$$(23)(13) = (132) = \sigma_{(23)}(4) = 6$$

$$(23)(123) = (12) = \sigma_{(23)}(5) = 2$$

$$(23)(132) = (13) = \sigma_{(23)}(6) = 4.$$

Thus $\sigma_{(23)} = (13)(25)(46)$. Since the permutation representation of a group is a homomorphism we know that the generators have the following property, $\sigma_{S_3} = \sigma_{\langle (12), (23) \rangle} = \langle (12)(45)(36), (13)(25)(46) \rangle$. □

Exercise (4.3.6 - Stefano Fochesatto). Assume G is a non-abelian group of order 15. Prove that $Z(G) = 1$. Use the fact that $\langle g \rangle \leq C_G(g)$ for all $g \in G$ to show that there is at most one possible class equation for G .

Proof: Suppose G is a non-abelian group of order 15. By Lagrange's Theorem we know that if G is non-abelian, then $|Z(G)| = 1, 3$ or 5 . If $|Z(G)| = 3$ then by Lagrange's Theorem $|G/Z(G)| = 5$ and therefore cyclic. By Exercise 3.1.36 it would then follow that G is cyclic and therefore abelian, a contradiction. Similarly if $|Z(G)| = 5$ then by Lagrange's Theorem $|G/Z(G)| = 3$ and therefore cyclic. Again it follows that G is cyclic and abelian, a contradiction. Therefore $Z(G) = 1$.

Suppose that $\langle g \rangle \leq C_G(g)$ for all $g \in G$. Note that $3|15$ and $5|15$ therefore by Cauchy's Theorem there exists an $a, b \in G$ such that $|a| = 3$ and $|b| = 5$. Since $\langle a \rangle \leq C_G(a) \leq G$ and $a \notin Z(G)$ with $|G : \langle a \rangle| = 5$, we get that $C_G(a) = \langle a \rangle$. Similarly since $\langle b \rangle \leq C_G(b) \leq G$ and $b \notin Z(G)$ with $|G : \langle b \rangle| = 3$, we get that $C_G(b) = \langle b \rangle$. Applying the class equation we know that,

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|,$$

For some conjugacy class representatives $g_1, g_2, \dots, g_r$. Note that each summand in $\sum_{i=1}^r |G : C_G(g_i)|$ must be a divisor of $|G| = 15$ So we get,

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)| = 1 + 3i + 5j,$$

where $i, j > 1$ since $\langle a \rangle, \langle b \rangle \leq G$. Note that this condition gives us that $i = 3$ and $j = 1$, since no other combination satisfies $14 = 3i + 5j$ with $i, j > 1$. $\square$